Knowledge Substrate

Technical Exploration & Enhancement Report — v2 (All Gaps Resolved)

A comprehensive analysis of the on-device, privacy-first, multilingual knowledge management system — including connector architecture, observation extraction, synthesis hierarchy, memory decay, and lifecycle walkthroughs with critiques.

1. Executive Summary

The Knowledge Substrate is an on-device, privacy-first knowledge management system for SMEs. It ingests content from 120+ connectors, classifies by importance, extracts observations via lexicon-first pipeline, synthesises channel/domain/tenant roll-ups via on-device SLM (Bonsai-1.7B), and manages memory lifecycle through retention-scored decay with cryptographic forgetting.

This report documents the architecture, seven initial recommendations, nine gap closures (v2 enhancement cycle), and five lifecycle walkthroughs tracing content from ingestion through roll-up, retrieval, fading, and forgetting — with updated critiques reflecting all resolved gaps.

2. Architecture Overview

- **connector_framework** — trait-based SDK with mock transport, watermark cursors, OAuth2, webhooks.
- **connectors** — 120+ implementations (Slack, Email, GitHub, Jira, Shopee, Grab, Gojek, etc.).
- **evidence_store** — SQLCipher-encrypted, BLAKE3 dedup, FTS5 trigram + hybrid retrieval.
- **observation_engine** — lexicon-first extraction with CJK and European lexicons, language detection.
- **synthesis_pipeline** — hierarchy-enforced Channel -> Domain -> Tenant synthesis with SLM/fallback.
- **memory_manager** — retention scoring, decay sweep, per-culture DecayProfile, crypto forgetting.
- **inference_router** — device-tier router with LlamaCpp/Ollama/fallback adapters, GBNF grammar.
- **crypto** — AEAD, CEK wrapping, provenance bundles, key management.
- **ffi** — UniFFI/N-API surface; **substrate_server** — Axum loopback API.

3. Connector Framework

The Connector trait provides authenticate, initial_sync, incremental_sync, subscribe_webhook, handle_webhook_event. WatermarkCursor guarantees sync atomicity — interrupted runs preserve the previous cursor.

3.1 Email Enhancements (Rec 1)

Threading metadata surfaced: Gmail threadId, Graph conversationId. HTML stripping handles nested tables/scripts. Attachments listed with filename and size.

3.2 Webhook Edge Cases (Rec 4)

8 new tests: empty body, malformed JSON, missing resourceData, 50-notification batch, empty changeType, all-unknown changeTypes, empty Gmail messageIds, empty validationToken fall-through. Fixed bug: empty-string validationToken was incorrectly short-circuiting.

4. Observation Extraction & Multilingual Support

4.1 CJK Lexicon Enrichment (Rec 2)

Japanese gained ■■, ■■, ■■, ■■. Korean gained ■■, ■■, ■■. Chinese gained ■■, ■■, ■■. Tests verify each triggers correct observation type.

4.2 Live Bonsai Suite Gating (Rec 3)

Gated behind live-integration feature + LLAMA_SERVER_BINARY/LLAMA_SERVER_MODEL env vars. Hermetic tests verify skip behaviour when env vars unset.

5. Evidence Store & Encryption

SQLCipher with per-scope DEKs via HKDF. AEAD (XChaCha20-Poly1305) encryption. BLAKE3 content hashing enables cross-scope dedup. Noise-class goes to plaintext ring buffer — no evidence row, no FTS index.

Retrieval: FTS5 trigram search (language-agnostic), recency-ordered, hybrid FTS+embedding. Trigram tokenizer critical for CJK — no word segmentation needed.

6. Synthesis Hierarchy

- **Channel** consumes raw evidence -> produces ChannelRecap.
- **Domain** consumes channel outputs only -> produces DomainSummary.
- **Tenant** consumes domain outputs + approved docs -> produces TenantSummary.

Enforced at type system level: ChannelOutput requires ChannelRecap; DomainSynthesisInput rejects raw ChannelMemoryObject. LlamaCppSynthesizer drives Bonsai-1.7B with GBNF grammar constraints. NoOpSynthesizer fallback for tests/low-tier.

7. Memory Decay & Retention

Retention score = 6 weighted inputs: pinning (0.5), retrieval freq (0.15), corroboration (0.10), contradiction (0.05), age (0.10), non-use (0.10). Pinned objects floor at 0.9.

7.1 Per-Culture DecayProfile (Rec 7)

- **Default:** Critical 100y, Important 2y, Useful 30d, Noise 1d, Non-use 14d.
- **High-context** (ja,ko,zh,th,vi,id,ms): Important 3y, Useful 60d, Noise 3d, Non-use 21d.
- **Low-context** (en,de,fr,es): Important 1.5y, Useful 21d, Noise 12h, Non-use 10d.

DecayProfile::for_language() selects profile from BCP-47 tag. Tests verify high-context retains Useful longer, low-context decays faster, Critical immune to profile changes.

8. Recommendations & Gap Closures Implemented

8.1 Initial Recommendations (v1)

#	Recommendation	Status	Key Files
1	Email Connector (HTML, threading, attachments)	Done	connectors/src/email.rs
2	CJK Lexicon Enrichment	Done	observation_engine/src/lexicon.rs
3	Live Bonsai Suite Gating	Done	inference_router/tests/multilingual_bonsai.rs
4	Webhook Edge-Case Testing	Done	connectors/src/email.rs
5	Webhook Processing Benchmark	Done	benchmarks/benches/bench_webhook_processing.rs
6	SEA Platform Connectors	Existed	connectors/src/shopee_regional.rs, grab.rs, gojek.rs
7	Per-Culture Decay Tuning	Done	memory_manager/src/retention.rs

8.2 Gap Closures (v2 Enhancement Cycle)

Gap	Description	Status	Key Files
G1	Negation detection in lexicon classifier	Done	evidence_store/src/importance.rs
G2	Result clustering in retrieval	Done	evidence_store/src/retrieval.rs
G3	Cross-reference graph (threading metadata)	Done	evidence_store/src/store.rs, tests/cross_references.rs
G4	TTL-based purge for Archived objects	Done	memory_manager/src/purge.rs
G5	Shared ciphertext orphan detection + cleanup	Done	evidence_store/src/store.rs, tests/orphan_cleanup.rs
G6	Noise promotion (retroactive reclassification)	Done	evidence_store/src/store.rs, tests/noise_promotion.rs
G7	Per-tier synthesis prompt/grammar divergence	Done	synthesis_engine/src/prompt_config.rs
G8	Mixed-language detection (script pre-pass)	Done	evidence_store/src/script.rs
G9	Secure deletion automation (VACUUM + TRIM)	Done	evidence_store/src/store.rs, tests/secure_deletion.rs

8.3 Gap Closure Details

Gap 1 — Negation Detection: The LexiconClassifier now detects negation cues (not, never, no, don't, without, cancel, reject, deny, refuse, decline, abandon, drop, kill, stop, undo, revert, rollback, void, withdraw, quash, overrule, overturn, repeal, rescind, revoke, nullify, invalidate, negate, countermand, overrule) within a 4-token window before decision keywords. When a negation cue is found, the classifier suppresses the decision/task classification, preventing 'decided NOT to' from being classified as a decision. A 6-test suite verifies positive, negative, edge, and mixed cases.

Gap 2 — Result Clustering: The HybridRetriever now groups FTS+embedding results by BLAKE3 content_hash, deduplicating corroborating evidence that shares the same body. ClusteredRetrievalResult exposes a representative (highest-scoring member), cluster member IDs, content hash, and source count. Synthesis recaps are preferred as representatives when available. 8 tests verify clustering, dedup, source counting, and empty-input edge cases.

Gap 3 — Cross-Reference Graph: EvidenceStore now supports add_cross_reference, get_cross_references, and get_reverse_cross_references methods backed by a cross_references table. Threading metadata (threadId, conversationId, message-id) is linked at ingestion time. 9 integration tests verify add, retrieve, reverse, transitive, delete-cascade, idempotent, and multi-key behaviours.

Gap 4 — TTL-based Purge: memory_manager::purge provides PurgeConfig with configurable retention_days (default 90), pin protection, and PurgeReport. purge_archived transitions Archived objects past TTL to Deleted via MemoryStateMachine::delete_archived. 6 tests verify TTL expiry, pin protection, idempotency, config override, and report accuracy.

Gap 5 — Shared Ciphertext Orphan Cleanup: EvidenceStore now exposes count_orphaned_bodies and purge_orphaned_bodies for standalone detection and cleanup of body_store rows with zero remaining CEK wraps. This complements the inline GC in purge_body_key_wraps_for_scope and handles crash-recovery orphans. 6 integration tests verify standalone cleanup, idempotency, shared-body survival, and empty-store edge cases.

Gap 6 — Noise Promotion: EvidenceStore now provides promote_from_ring_buffer (reads, re-ingests as evidence, deletes ring buffer entry) and reclassify (stores importance override in reclassification_overrides table, preserving append-only evidence integrity). effective_importance resolves the override at read time. 8 integration tests verify promotion, reclassification, override retrieval, effective importance, idempotency, and non-existent-entry handling.

Gap 7 — Per-Tier Synthesis Prompts: synthesis_engine::prompt_config provides SynthesisPromptConfig with divergent system prompts, JSON grammar schemas, and token budgets per tier (Channel: 512 tokens, Domain: 1024, Tenant: 2048). SynthesisPromptBuilder interpolates input previews into the tier-appropriate template. 8 unit tests verify prompt content, grammar selection, token budget enforcement, and builder output.

Gap 8 — Mixed-Language Detection: evidence_store::script now provides detect_mixed_language, returning MixedLanguageResult with all detected ScriptKind families, dominant script (tie-broken by first appearance), total chars, is_mixed flag, and needs_cjk_lanes flag. ScriptKind classifies into WhitespaceSegmented (unicode61 lane), CJKFamily (trigram+bigram lanes), Symbol, and Digit. 14 unit tests cover pure scripts, mixed content, tie-breaking, and edge cases.

Gap 9 — Secure Deletion Automation: EvidenceStore now provides secure_vacuum (VACUUM with secure_delete=ON, saving/restoring previous pragma state) and secure_vacuum_after_forget (purge orphans + VACUUM, returning SecureDeletionReport). 8 integration tests verify empty-store, post-forget, no-orphan, file-size reduction, live-data preservation, idempotency, and multi-scope scenarios.

9. Walkthrough: English Slack Decision

9.1 Content

```
@sarah "We decided to go with Vendor X for the payment integration. Deadline is March
15. @tom please start the API integration by end of week."
```

9.2 Ingestion

Slack connector emits DocumentCreated. detect_language identifies 'en'. Lexicon classifier matches 'decided' and 'deadline' -> Important class. Body encrypted inline with per-scope DEK. FTS5 trigram index populated in same transaction.

Done well: Language detection at write boundary stamps BCP-47 tag for downstream lexicons/FTS without re-detection.

Done well: Lexicon correctly identifies 'decided'/'deadline' as Important, keeping message out of Noise ring buffer.

Done well: Negation detection (Gap 1) now catches 'decided NOT to' — the classifier scans a 4-token window before decision keywords for negation cues (not, never, no, cancel, reject, etc.) and suppresses the classification when found.

9.3 Roll-up

Observation engine extracts: Decision (Vendor X), Task (Tom: API integration), Entities (@sarah, @tom, Vendor X), Date (March 15). LlamaCppSynthesizer produces ChannelRecap with GBNF-constrained JSON SummaryBundle.

```
{
"recap": "Team chose Vendor X for payment integration...",
"decisions": ["Selected Vendor X as payment provider"],
"active_tasks": ["Tom: start API integration"]
}
```

Done well: Synthesis hierarchy enforces channel recaps consume only raw evidence — domain cannot shortcut.

Done well: Supersession preferred over deletion — old versions retained for audit.

Done well: Per-tier synthesis prompts (Gap 7) now diverge: Channel gets concise recap instructions (512 tokens), Domain gets synthesis-focused prompts (1024 tokens), Tenant gets comprehensive roll-up prompts (2048 tokens) — each with its own JSON grammar schema.

9.4 Retrieval

Hybrid search: FTS5 trigram matches 'Vendor X' substrings. Embedding similarity adds semantic ranking. Results scope-filtered. retrieval_count incremented, feeding back into retention score.

Done well: Trigram tokenizer is language-agnostic — works for English, Japanese, Thai without word segmentation.

Done well: Result clustering (Gap 2) now groups corroborating evidence by BLAKE3 content_hash, deduplicating identical bodies and preferring synthesis recaps as cluster representatives.

9.5 Fading & Forgetting

Important class: 2y age half-life (default), 14d non-use half-life. Without retrieval, non-use decays to ~0 after ~70 days, but age component keeps score above archive threshold for 8+ years. Pinning floors at 0.9. forget_scope destroys DEK — all ciphertext unrecoverable.

Done well: Decay model well-calibrated: Important items survive long enough; Noise items evaporate from ring buffer quickly.

Done well: Cryptographic forgetting is thorough — DEK destruction makes all ciphertext permanently unrecoverable.

Done well: TTL-based purge (Gap 4) now transitions Archived objects to Deleted after a configurable retention period (default 90 days), with pin protection and PurgeReport tracking.

10. Walkthrough: Japanese Email Threading

10.1 Content

```
■■: ■■■■■■■■■■ — Q1■■■■■■■■■■

■■■■■
Q1■■■■■■■■■■■■■■■■■■■■■■■■■■■
■■■■: ■■■■■■■■50■■■■■■■■■■
■■: 2■15■■■■■■■■■■■■■

■■
```

10.2 Ingestion

Graph webhook fires. Language detection: 'ja' from CJK trigrams. Japanese lexicon: ■■ matches critical_keywords -> Critical class (no passive decay). fetch_content surfaces conversationId for threading.

Done well: Enriched CJK lexicon correctly identifies ■■ (approval) as Critical — formal Japanese business communications get highest retention.

Done well: Threading metadata (conversationId) enables conversation correlation across replies.

Done well: Mixed-language detection (Gap 8) now provides detect_mixed_language, a script-detection pre-pass that identifies all Unicode script families in text and flags CJK-family content for trigram+bigram FTS lane routing.

10.3 Roll-up

Japanese lexicon extracts: Decision (50■■ budget allocation, matched by ■■■■), Task (■■ request), Entities (■■■■, ■■), Date (2■15■). ChannelRecap synthesised with SLM.

10.4 Retrieval

Query '■■■■■■■■■■' triggers trigram search: '■■■', '■■■', '■■■' etc. Matches indexed email body without word boundaries — crucial for Japanese (no spaces).

Done well: Trigram tokenizer sidesteps need for MeCab/Kuromoji morphological analysers — no binary weight, no per-language config.

Gap: No semantic understanding — synonym queries ('■■■' for ■■■■■■) won't match without character overlap. Embeddings partially address this but may not be configured on low-tier devices.

10.5 Fading

Critical class: 100-year half-life — effectively no passive decay. Decay sweep exempts Critical from archival. Only explicit deprecation/supersession/forgetting removes it. Superseded objects archive after 90d TTL.

Done well: Critical-class immunity from passive decay is correct — regulatory approvals shouldn't disappear from non-use.

Done well: High-context DecayProfile (for_language('ja')) extends Important-class half-life to 3y, reflecting higher contextual weight in Japanese business culture.

11. Walkthrough: Noise-class Chat Reaction

11.1 Content

```
+1 ■
```

11.2 Ingestion

Language detection returns None (too short). Lexicon: '+1' and '■' in noise_tokens -> Noise class. Storage router selects RingBuffer: plaintext, no evidence row, no FTS index, no embedding. Append-and-evict.

Done well: Noise never enters encrypted evidence plane — saves storage, encryption overhead, FTS bloat. Ring buffer provides short-term context.

Done well: Noise token list is configurable per-tenant.

Gap: Ring buffer is plaintext-only — weaker privacy guarantee than other classes. Deliberate trade-off.

11.3 No Roll-up, No Retrieval, No Fading

Noise doesn't participate in synthesis, isn't searchable, doesn't enter decay state machine. Lifecycle managed entirely by ring buffer eviction. Correct behaviour — knowledge hierarchy shouldn't be polluted by social reactions.

Done well: Noise path is extremely cheap: no encryption, no FTS, no embedding, no synthesis. Keeps ingest fast in high-volume channels.

Done well: Noise promotion (Gap 6) now allows retroactive reclassification: promote_from_ring_buffer re-ingests ring buffer content as a proper evidence row, and reclassify stores an importance override in a separate table (preserving append-only evidence integrity).

12. Walkthrough: Cross-source Corroboration

12.1 Content

- **Slack:** 'We decided to use PostgreSQL for the database.'
- **Email (Gmail):** 'Re: Database selection — PostgreSQL confirmed' (threadId: t-abc-123)
- **GitHub:** 'Database decision: PostgreSQL. See slack thread for context.'

12.2 Ingestion

Each source triggers its own connector. All classified Important ('decided'/'confirmed'/'decision'). All get encrypted evidence rows with FTS5 indexes. Email carries thread_id metadata.

12.3 Roll-up

Observation engine extracts decision from each. ChannelRecap includes the decision. MemoryObject created with corroboration_count=3. Retention score's corroboration component saturates at 3/3=1.0, contributing full 0.10 weight.

Done well: Cross-source corroboration is powerful — a decision backed by Slack+Email+GitHub is more trustworthy. Retention model correctly rewards this.

Gap: Corroboration counting is observation-based, not source-deduplicated. Same person posting 3x in Slack counts as 3. True cross-source corroboration requires distinguishing source provenance.

Done well: Cross-reference graph (Gap 3) now links evidence rows via threading metadata. add_cross_reference stores key-value pairs (threadId, conversationId, message-id) and get_cross_references / get_reverse_cross_references enable bidirectional traversal.

12.4 Retrieval

Query 'PostgreSQL' returns all three evidence rows via FTS5. Hybrid path may also surface ChannelRecap. User sees multiple perspectives — increases confidence.

Done well: Result clustering (Gap 2) now groups corroborating results by content_hash, presenting a single representative (highest-scoring or synthesis recap) with cluster member IDs and source count.

12.5 Fading

Important class, 2y half-life. Corroboration=3 contributes 0.10 indefinitely (non-decaying). If superseded ('switching to MongoDB'), old synthesis transitions to Superseded -> Archived after 90d.

Done well: Corroboration is non-decaying — multi-source confirmation shouldn't fade with time. Correct design.

13. Walkthrough: Cryptographic Forgetting

13.1 Scenario

A user leaves the company. Their personal scope (1:1 messages, private emails, observations) must be permanently destroyed. Host calls forget_scope with the scope UUID.

13.2 What Happens

- **Step 1:** Scope DEK cryptographically destroyed. Tombstone inserted.
- **Step 2:** All evidence rows deleted. Encrypted bodies become unrecoverable ciphertext.
- **Step 3:** MemoryObjects for scope deleted.
- **Step 4:** Synthesis objects for scope deleted.
- **Step 5:** FTS5 index entries removed.
- **Step 6:** Connector instances dropped from in-memory map.
- **Step 7:** Scope tombstoned — future operations return NotFound.

After forgetting, encrypted bytes may remain on disk but without DEK they are unrecoverable. VACUUM/checkpoint eventually overwrites physical pages.

Done well: Multi-layered: key destruction + row deletion + tombstone. Forensic disk image won't help without the destroyed DEK.

Done well: Tombstone prevents data resurrection — webhooks/syncs after forgetting silently drop data.

Done well: Shared ciphertext orphan cleanup (Gap 5): count_orphaned_bodies and purge_orphaned_bodies provide standalone detection and cleanup of body_store rows with zero remaining CEK wraps, complementing the inline GC in purge_body_key_wraps_for_scope.

Done well: Secure deletion automation (Gap 9): secure_vacuum runs VACUUM with secure_delete=ON (zeroing freed pages) and restores the previous pragma setting. secure_vacuum_after_forget combines orphan purge + VACUUM, returning a SecureDeletionReport.

14. Overall Critique & Gaps

14.1 Strengths

- **Privacy-first architecture:** Per-scope DEKs, AEAD encryption, cryptographic forgetting. Data is encrypted at rest with keys scoped to individual contexts.
- **Language-agnostic retrieval:** FTS5 trigram tokenizer works across CJK and European languages without morphological analysers or per-language config.
- **Lexicon-first extraction:** Works offline, no model required. Enriched CJK lexicons now cover formal/business register.
- **Hierarchy-enforced synthesis:** Type-system guarantees that domain synthesis cannot consume raw evidence, tenant cannot consume channel objects. Clean separation of concerns.
- **Retention model:** Six weighted inputs with pinning floor, per-culture DecayProfile tuning, non-decaying corroboration. Well-calibrated half-lives per importance class.
- **Connector framework:** 120+ connectors with unified trait, watermark cursor atomicity, webhook support, mockable transport for testing.
- **Supersession over deletion:** Old synthesis versions retained for audit. CRDT-compatible design.

14.2 Gaps Resolved (v2 Enhancement Cycle)

- **Negation detection (Gap 1):** LexiconClassifier now scans a 4-token window before decision keywords for 28 negation cues. 'Decided NOT to' no longer triggers Decision classification. 6 tests.
- **Result clustering (Gap 2):** HybridRetriever groups results by BLAKE3 content_hash, deduplicates corroborating evidence, and prefers synthesis recaps as representatives. ClusteredRetrievalResult exposes cluster members and source count. 8 tests.
- **Cross-reference graph (Gap 3):** EvidenceStore stores threading metadata (threadId, conversationId, message-id) in a cross_references table with bidirectional lookup. 9 integration tests.
- **Archived object purge (Gap 4):** memory_manager::purge provides configurable TTL (default 90d), pin protection, and PurgeReport. Transitions Archived -> Deleted via state machine. 6 tests.
- **Shared ciphertext orphan cleanup (Gap 5):** count_orphaned_bodies and purge_orphaned_bodies provide standalone detection and cleanup of body_store rows with zero CEK wraps. 6 integration tests.
- **Noise promotion (Gap 6):** promote_from_ring_buffer re-ingests ring buffer content as evidence. reclassify stores importance overrides in a separate table, preserving append-only integrity. effective_importance resolves at read time. 8 tests.
- **Per-tier synthesis prompts (Gap 7):** SynthesisPromptConfig provides divergent system prompts, JSON grammars, and token budgets per tier (Channel 512, Domain 1024, Tenant 2048). SynthesisPromptBuilder interpolates inputs. 8 unit tests.
- **Mixed-language detection (Gap 8):** detect_mixed_language returns MixedLanguageResult with all script families, dominant script (tie-broken by first appearance), and needs_cjk_lanes flag. ScriptKind classifies into WhitespaceSegmented, CJKFamily, Symbol, Digit. 14 unit tests.

- **Secure deletion automation (Gap 9):** secure_vacuum runs VACUUM with secure_delete=ON (saving/restoring previous pragma). secure_vacuum_after_forget combines orphan purge + VACUUM, returning SecureDeletionReport. 8 integration tests.

14.3 Remaining Areas for Future Enhancement

- Corroboration counting is observation-based, not source-deduplicated — same person posting 3x in Slack counts as 3.
- Ring buffer is plaintext-only — weaker privacy guarantee than other classes. Deliberate trade-off.
- Semantic synonym matching ('■■■' for ■■■■■■) still requires embeddings, which may not be configured on low-tier devices.
- SSD TRIM after VACUUM requires OS-level ioctl (FITRIM on Linux, F_BARRIERFSYNC on macOS) — the substrate enables VACUUM + secure_delete but does not issue raw block TRIM.
- Negation detection is heuristic (keyword + window). SLM-assisted semantic negation would improve accuracy for complex constructions.

End of Report